Narges Rezaie

Senior Bioinformatics Engineer | Computational Biologist | AI/ML Engineer

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Summary

Bioinformatics Engineer & **Ph.D. Computational Biologist** (**UC Irvine**) with **7+ years** of experience. Expert in building **scalable pipelines** for **long/short-read sequencing** (**ONT, PacBio, Illumina**), **bacterial genomics**, and **single-cell multi-omics**. Specialist in **Nextflow, Snakemake,** and **AWS cloud-native** deployments. Pioneering **LLM-based agentic systems** to automate **bioinformatics workflows** and software tooling.

Work Experience

Bioinformatics Pipeline Engineer — **Plasmidsaurus**
CA, USA — *Dec 2024 – Present*

- **Own the end-to-end development** and **productionization** of bioinformatics pipelines, serving as the **primary lead** for **bacterial production workflows** and RNA-seq product support.
- **Architect scalable workflow systems** using **Nextflow** and **Snakemake** across **AWS Batch** and HPC environments, ensuring **reproducible results** for high-throughput sequencing data.
- **In charge of the development and launch** of **Illumina metagenomics** and **small variant calling** services; **benchmarked and integrated** advanced tools for **taxonomic profiling** and **host-read removal** to optimize **turnaround time** and result accuracy.
- **Optimize pipeline performance** by resolving complex **biological, sequencing, and infrastructure edge cases** from initial data ingestion to the final customer deliverable.
- **Drive platform evolution** as a **full-stack contributor**, developing **backend logic** and **frontend features** that enhance **data visualization** and the seamless delivery of genomic results.
- **Pioneer the integration** of **LLM and Agent-based systems** to automate **bioinformatics workflows**, streamline complex data analysis, and elevate **internal developer tooling**.
- **Collaborate cross-functionally** with scientists, engineers, and product teams to **translate novel computational methods** into **robust, customer-facing** bioinformatics products.

Research Experience

Postdoctoral Fellow — **Mortazavi Lab, UCI**
CA, USA — *March 2024 – Dec 2024*

- Optimized bioinformatic algorithms for NGS datasets including short/long-read RNA-seq (bulk and single-cell), Multiome, Perturb-seq, and DNA methylation data.
- Characterized the impact of genomic variants and regulatory elements on phenotypes (Nature, 2024).
- **Gene program evaluation (GitHub)**: Implemented a flexible framework to evaluate the plausibility of gene programs inferred from single-cell genomic data.

Graduate Researcher — **Mortazavi Lab, UCI**
CA, USA — *Sept 2019 – March 2024*

- **Model-AD (GitHub)**: Analyzed diverse genomics datasets to characterize animal models for Alzheimer's and developed predictive models for therapeutic discovery.
- **IGVF** & **ENCODE4**: Investigated how genomic variation affects function using single-cell data and profiled full-length transcriptomes in humans and mice.
- **PyWGCNA (GitHub)**: Developed a Python package for weighted correlation network analysis, enabling comparison across datasets and functional module enrichment.
- **Topyfic (GitHub)**: Built an unsupervised ML tool using Latent Dirichlet Allocation (LDA) to identify and annotate gene programs from single-cell data.
- Mentored undergraduate students in bioinformatics programming and reproducible pipeline development.

Education

PhD in Mathematical, Computational, and Systems Biology **USA**
University of California, Irvine *2019–2024*

- **Thesis**: Computational approach to characterize gene dynamics using bulk and single-nucleus RNA sequencing. **GPA**: 3.98/4.0

B.S. in Information Technology in Computer Engineering **Iran**
Sharif University of Technology *2014–2018*

Skills

- **Bioinformatics:** Genome assembly (Hifiasm, Flye), Variant calling (Clair3), Metagenomics, Bulk/Single-cell RNA-seq, Multi-omics Integration, QC/Benchmarking.
- **Programming:** Python (Expert), R, Bash, SQL, C/C++, FastAPI, Pydantic, Pytest, Java.
- **Cloud & DevOps:** AWS (Batch, S3, EC2, Lambda), Nextflow (DSL2), Snakemake, Docker, GitHub Actions (CI/CD), SLURM, Git, Linux.
- **AI/Machine Learning:** LLM Agents, PyTorch, Scikit-learn, Unsupervised Learning (LDA), Statistical Modeling.
- **Python Stack:** Pandas, Polars, NumPy, SciPy, AnnData, Biopython, Vector Databases.

Selected Publications

- Identification of robust cellular programs using reproducible LDA that impact sex-specific disease progression in different genotypes of a mouse model of AD. **N Rezaie**, E Rebboah, et al. *Briefings in Bioinformatics* (2025).
- The ENCODE mouse postnatal developmental time course identifies regulatory programs of cell types and cell states. E Rebboah, **N Rezaie**, et al. *Cell Genomics* (2025).
- Long-read sequencing transcriptome quantification with lr-kallisto. RK Loving, ..., **N Rezaie**, ..., L Pachter. *PLoS Computational Biology* 21(12): e1013692 (2025).
- BIN1^{K358R} suppresses glial response to plaques in mouse model of Alzheimer's disease. LF Garcia-Agudo, ..., **N Rezaie**, ..., KN Green. *Alzheimer's & Dementia* (2025).
- The Abca7^{V1613M} variant reduces Aβ generation, plaque load, and neuronal damage. CA Butler, ..., **N Rezaie**, ..., GR MacGregor, KN Green. *Alzheimer's & Dementia* (2025).
- The ENCODE4 long-read RNA-seq collection reveals distinct classes of transcript structure diversity. F Reese, ..., **N Rezaie**, et al. *Nature Methods* (2024).
- Deciphering the impact of genomic variation on function. IGVF Consortium. *Nature* (2024).
- A Trem2^{R47H} mouse model without cryptic splicing drives age-and disease-dependent tissue damage and synaptic loss in response to plaques. KM Tran, ..., **N Rezaie**, et al. *Molecular Neurodegeneration* (2023).
- PyWGCNA: A Python package for weighted gene co-expression network analysis. **N Rezaie**, F Reese, A Mortazavi. *Bioinformatics* (2023).
- Somatic point mutations are enriched in non-coding RNAs with possible regulatory function in breast cancer. **N Rezaie**, ..., H Alinejad-Rokny. *Communications Biology* (2022).

Selected Talks and Posters

- Alzheimer's Association International Conference 2024. *Poster*: "5xFAD/Abi3^{S209F} mice display distinct effects on microglia dynamics."
- PARSE Biosciences 2023. *Invited Speaker*: "Identifying Distinct Cellular Programs Using Topyfic."
- Network Biology 2023 (CHSL). *Poster*: "Topyfic identified robust cellular programs using reproducible LDA."
- Alzheimer's & Dementia 2022 (AAIC). *Poster*: "Bulk and single-cell gene expression network analysis using PyWGCNA."
- Neuroscience 2020 (SFN). *Poster*: "Bulk and single-nucleus analysis of the 3xTgAD mouse model."

References

Available upon request.